Cambodia Academy of Digital Technology
Course: Software Engineering

Research Assignment: Investigating the Role, Impact, and Relevance of UML in Software Engineering in Modern Software Engineering

Name: Chey Naryvety

ID: IDTB110199

1 Why UML Exists

1. Why was UML created in Software Engineering

   UML was created to define a standard way to visualize, design, document, and communicate system design

   - What problems was it trying to solve?

- Complex applications need collaboration and planning from multiple teams, so a clear and concise way of communication is required.
- Business stakeholders may not understand code, so UML helps explain system requirements, functionalities, and processes in a simple visual way.
- It saves time by allowing teams to visualize workflows, user interactions, and system structure before actual development.
   - Were developers struggling before UML?

Yes, before UML, developers often used many different modeling methods and diagrams and they relied on written document and source code, making collaboration difficult. As software projects became larger and more complex, developers and teams needed a unified and standardized modeling.

2. Could software engineering function effectively without UML Explain your reason.

Software engineering can still function without UML when it comes to small projects or startups. But larger and complex project/system will become harder to manage without proper modeling as requirements may be misunderstood, developers may build system differently, poor communication, hard to maintenance and difficult to scale. Therefore, UML is not always needed but it is always better to have proper modeling approach.

3. Why do engineers model systems before implementation? Is coding immediately always a good approach? Explain!

Engineers model systems before implementation because it helps plan and identify the problems/issues early on while also allowing teams to understand system structure, workflow, and relationships before implementing begins. It acts as a blueprint, without model systems, developers may misunderstand requirements, system architecture might become disorganized, fixing mistakes later become hard, complicated, and expensive, team member may implement inconsistent solutions etc.

2 Impact of UML

1. How does UML improve communication among stakeholders? (Developer, Manager, Client, Tester)

   UML provides a shared visual language:

   - **Developers** - understand system structure and logic
   - **Managers** – track project structure and progress
   - **Clients** – visualize system behaviour without technical knowledge
   - **Testers** - derive test cases from diagrams and understand workflows

   Instead of long technical documents, UML diagrams make system ideas visual and easier to understand across roles.

2. How Does UML Reduce Software Development Risks?
   Example:

   - Misunderstood requirements

     Use case diagrams help clarify user needs and system functions early
   - Wrong architecture

     Class and component diagrams help design proper system structures before coding
   - Duplicated modules

     Modelling allows developers to identify repeated functions and avoid redundancy
   - Late project delivery

     Clear planning improves task organization and reduces development confusion
   - Costly redesign

     Detecting design problems early prevents expensive modifications later in development

3. How does UML improve the maintainability of software Imagine another developer joins after 2 years. How does UML help?

   If another developer joins after 2 years

   UML helps them:
   - Understand system structure quickly
   - See relationships between components
   - Identify dependencies without reading all code
   - Reduce time needed to onboard

   Without UML, they must rely only on code, which can be:
   - Complex
   - Poorly documented
   - Hard to reverse-engineer

4. How does UML support scalability?
   Can a system grow more easily when properly modeled?

   Yes, a proper modeled system can grow more easily.
   UML helps scalability by:
   - Designing modular architectures
   - Separating system components clearly
   - Defining relationships between modules
   - Making future expansion easier

   For example, in an ecommerce system, UML can help developers add online payment system, delivery tracking, customer reviews, mobile applications without redesigning the entire system.

3 If UML Does Not Exist

1. What would happen if software projects were developed without UML or any modeling approach?
   Possible angles:
   - Communication breakdown

     Different team members may interpret requirements differently
   - Requirement confusion

     Clients and developers may not share the same understanding of the system
   - Messy system structure

     Without planning, code structure may become unorganized and difficult to maintain
   - Integration failure

     Different modules may not work together properly because interfaces were not planned

2. Which Projects can survive without UML? Why?

    - Small apps?

      Simple applications with one or two developers may survive without UML because the system is small and easier to understand

    - Startup MVP?

      Startups often prioritize speed over documentation. Basic planning may be enough during early development

    - Enterprise banking system?

      These systems usually require UML because they are highly complex, security-sensitive, and involve many teams

    - Hospital System?

      Hospital systems also require UML because errors can affect patient safety, medical records, and system reliability

4 Reflection

Do you believe UML is still important in Agile/AI-assisted modern software development? why or why not?

Yes, I think UML is still important even in Agile and AI-assisted development.
Agile development focuses on flexibility and fast iteration, but teams still need clear communication and system understanding. UML diagrams can be simplified and updated continuously during Agile development.
AI tools can generate code faster, but AI still depends on correct system design and clear requirements. Poor architecture or misunderstood requirements cannot always be solved by AI.
UML remains valuable because it:

- Improves communication
- Helps organize complex systems
- Supports teamwork
- Reduces development risks
- Makes systems easier to maintain and scale

Although modern teams may use UML less formally than before, modeling concepts are still highly relevant in software engineering today.